

Mathematical Methods in Uncertainty Quantification (MMUQ) – L02

Faizan Anwar (faizan.anwar@tum.de)

Technical University of Munich

School of Engineering and Design

Chair of Hydrology and River Basin Management (Prof. Disse)

MMUQ L02 – First Assignment and Model Description

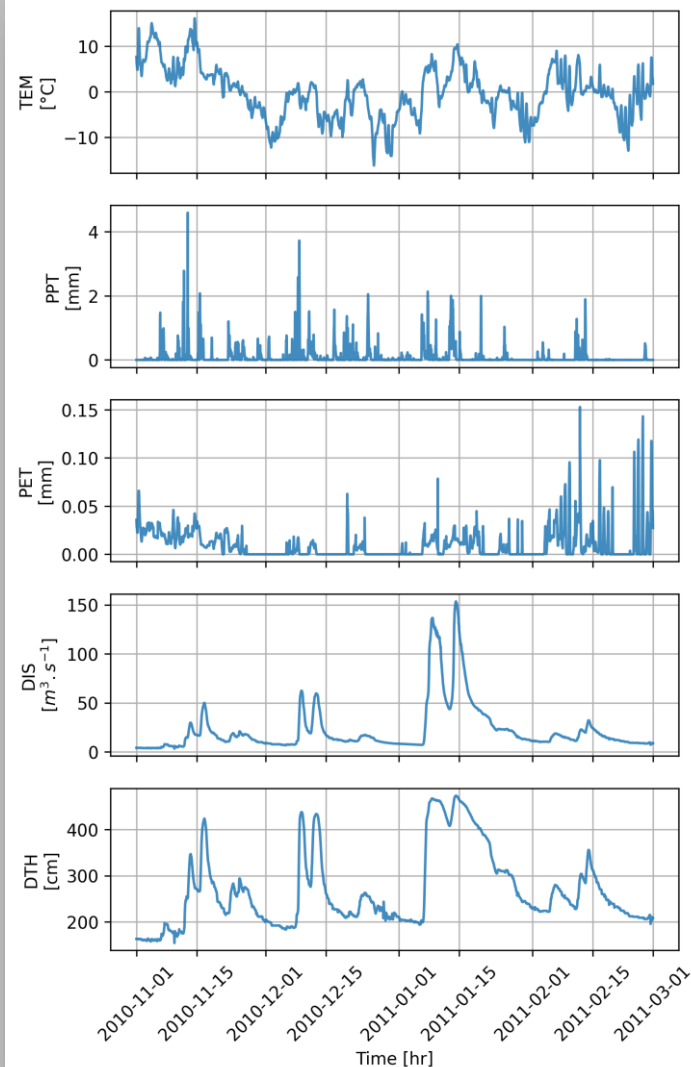
- Assignment
- Tasks
- HBV001a

Assignment – 1

- Global model parameter optimization using Differential Evolution
- High flow event
- Get a feeling of how model parameter optimization works
- Various objective functions
- Reference: Long-term lumped projections of groundwater balances in the face of limited data, Ejaz 2024, Ph.D. Thesis

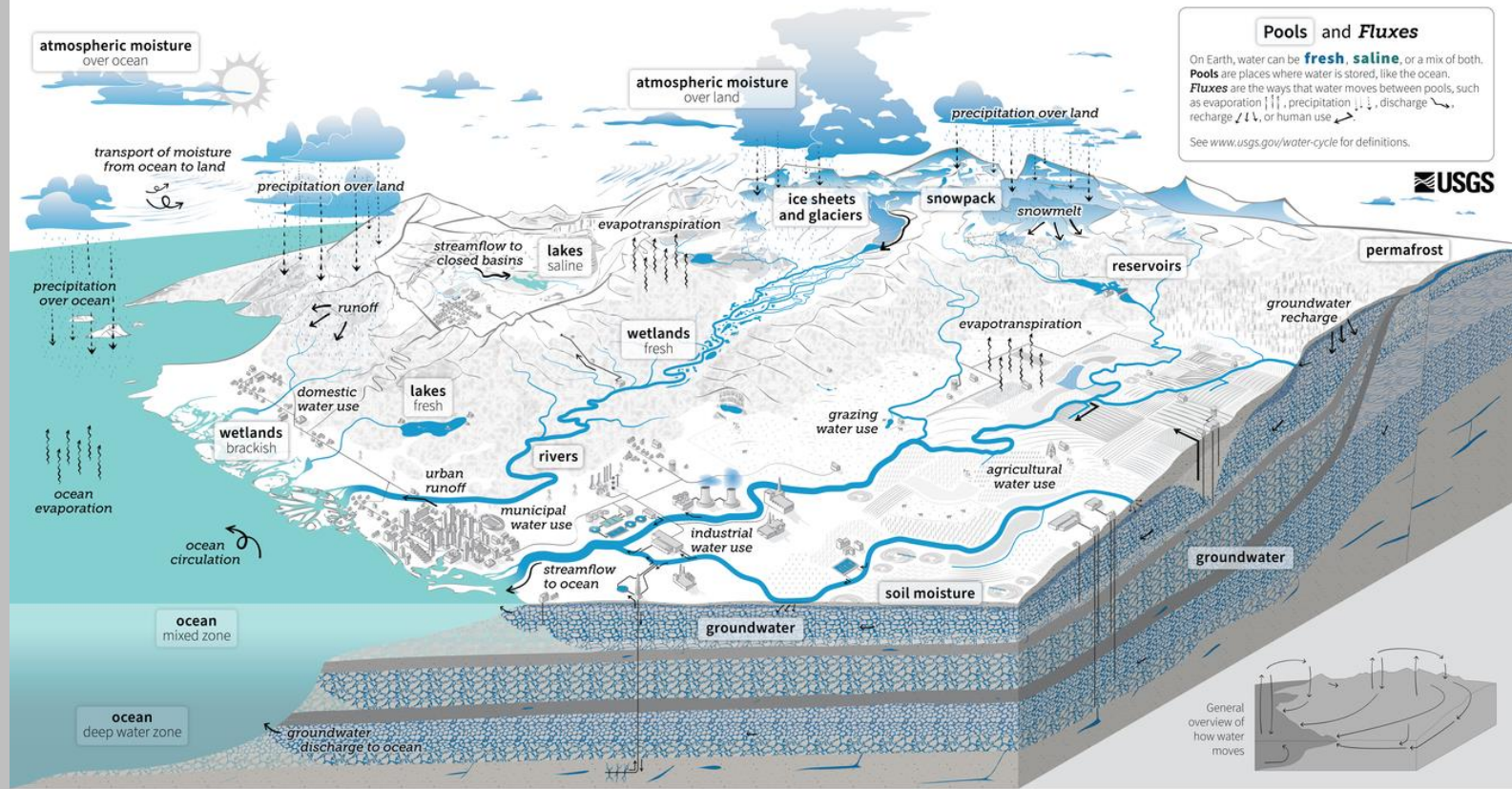
Assignment – 1, Problem

- Given
 - PPT time series
 - TEM time series
 - PET time series
 - Observed river flows (DISO)
 - Catchment area
- Lumped HBV001a
- Unknown model parameters
- Fit model to DISO
 - Changing parameters
 - via an optimization scheme



Assignment – 1, Tasks

- Implement various efficiency metrics
 - NSE, LnNSE, PBias...
- Implement the objective function (OF)
 - Accepts model parameters
 - Accepts model and related data
 - Runs model
 - Computes efficiency metrics
 - Returns objective function value (OFV)
- Implement the optimization scheme
- Obtain best performing parameter vector
- Estimate the least number of processes required
 - Similar overall efficiency
 - Makes physical sense
- Plot internal model variables
 - With all processes turned on
 - With some turned off
- Plot convergence of the OFVs
- Plot parameter vs. OFVs (each)
 - Can you tell which ones are sensitive?



The Water Cycle

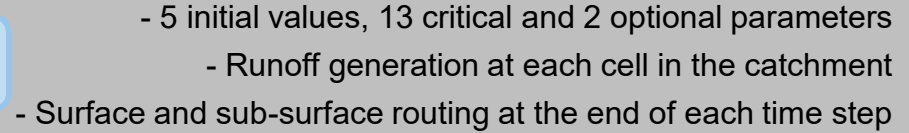
The water cycle describes where water is found on Earth and how it moves. Water can be stored in the atmosphere, on Earth's surface, or below the ground. It can be in a liquid, solid, or gaseous state. Water moves between the places it is stored at large scales and at very small scales. Water moves naturally and because of human interaction, both of which affect where water is stored, how it moves, and how clean it is.

Liquid water can be fresh, saline (salty), or a mix (brackish). Ninety-six percent of all water is saline and stored in **oceans**. Places like the ocean, where water is stored, are called **pools**. On land, saline water is stored in **saline lakes**, whereas fresh water is stored in liquid form in **freshwater lakes**, artificial **reservoirs**, **rivers**, **wetlands**, and in soil as **soil moisture**. Deeper underground, liquid water is stored as **groundwater** in aquifers, within the cracks and pores of rock. The solid, frozen form of water is stored in **ice sheets**, **glaciers**, and **snowpack** at high elevations or near the Earth's poles. Frozen water is also found in the soil as **permafrost**. Water vapor, the gaseous form of water, is stored as **atmospheric moisture** over the ocean and land.

As it moves, water can transform into a liquid, a solid, or a gas. The different ways in which water moves between pools are known as **fluxes**. **Circulation** mixes water in the oceans and transports water vapor in the atmosphere. Water moves between the atmosphere and the Earth's surface through **evaporation**, **evapotranspiration**, and **precipitation**. Water moves across the land surface through **snowmelt**, **runoff**, and **streamflow**. Through infiltration and **groundwater recharge**, water moves into the ground. When underground, groundwater flows within aquifers and can return to the surface through **springs** or from natural **groundwater discharge** into rivers and oceans.

Humans alter the water cycle. We redirect rivers, build dams to store water, and drain water from wetlands for development. We use water from rivers, lakes, reservoirs, and groundwater aquifers. We use that water (1) to supply our **homes and communities**; (2) for **agricultural** irrigation and **grazing** livestock; and (3) in **industrial** activities like thermoelectric power generation, mining, and aquaculture. The amount of available water depends on how much water is in each pool (water quantity). Water availability also depends on when and how fast water moves (water timing), how much water is used (water use), and how clean the water is (water quality).

Human activities affect **water quality**. In agricultural and urban areas, irrigation and precipitation wash fertilizers and pesticides into rivers and groundwater. Power plants and factories return heated and contaminated water to rivers. Runoff carries chemicals, sediment, and sewage into rivers and lakes. Downstream from these types of sources, contaminated water can cause harmful algal blooms, spread diseases, and harm habitats. **Climate change** is also affecting the water cycle. It affects water quality, quantity, timing, and use. Climate change is also causing ocean acidification, sea level rise, and extreme weather. Understanding these impacts can allow progress toward sustainable water use.



HBV001a – Snow Module

- Given precipitation (PPT)
- Given temperature (TEM)
- Snow (SNW) when too cold
 - Threshold temperature (TT)
 - SNW depth increases
- Melt (MLT) when too warm
 - Degree-day (AMF)
 - Liquid PPT
 - SNW depth decreases
- Rain-on-Snow
 - Optional (PMF)

$$MLT = \begin{cases} (TEM - TT) \times (AMF + (PPT \times PMF)), & \text{if } TEM > TT \\ 0, & \text{else} \end{cases}$$

$$MLT = MIN(SNW_{DTH}, MLT)$$

$$RNF_{POT} = \begin{cases} MLT + PPT, & \text{if } TEM > TT \\ 0, & \text{else} \end{cases}$$

$$SNW_{DTH} = \begin{cases} SNW_{DTH} - MLT, & \text{if } TEM > TT \\ SNW_{DTH} + PPT, & \text{else} \end{cases}$$

HBV001a – Soil Module

- Given PPT/MLT and PET
- Single layer (SOL)
 - Runoff (RNF) generation
 - Infiltration (ILN) and saturation excess
 - Via Shape Parameter (β)
 - Field capacity (FCY, > 0)
 - Evapotranspiration (ETN)
 - Permanent wilting point (PWP)
 - Only when TEM > 0 and SNW_{DTH} = 0

$$RNF = \left(\frac{SOL_{DTH}}{FCY} \right)^{\beta} \times RNF_{POT}$$

$$ILN = RNF_{POT} - RNF$$

$$SOL_{DTH} = SOL_{DTH} + ILN$$

$$RNF = RNF + MAX(SOL_{DTH} - FCY, 0)$$

$$SOL_{DTH} = MIN(SOL_{DTH}, FCY)$$

$$ILN = RNF_{POT} - RNF$$

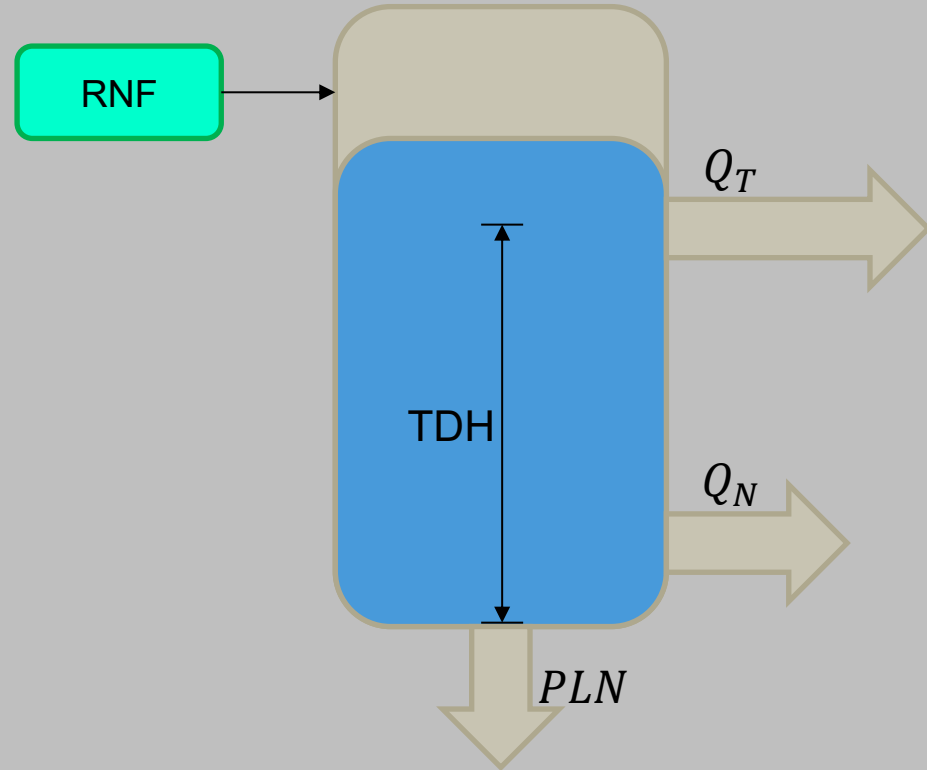
$$ETN = \frac{SOL_{DTH}}{PWP} \times PET$$

$$ETN = MIN(ETN, SOL_{DTH})$$

$$SOL_{DTH} = SOL_{DTH} - ETN$$

HBV001a – Point Runoff Generation (Surface fast flow)

- Given RNF from SOL
- Upper reservoir (URR)
 - Water split ratio (WSR, optional)
 - Faster response (NDR, TDR)
 - Two outlets (non-linear)
 - Surface (Q_T) and interflow (Q_N)
 - Percolation (PLN)
 - Active occasionally



HBV001a – Point Runoff Generation (Surface fast flow)

- Given RNF from SOL
- Upper reservoir (URR)
 - Water split ratio (WSR, optional)
 - Faster response (NDR, TDR)
 - Two outlets (non-linear)
 - Surface (Q_T) and interflow (Q_N)
 - Percolation (PLN)
 - Active occasionally

$$URR_{DTH} = URR_{DTH} + ((1 - WSR) \times RNF)$$

$$Q_T = (URR_{DTH} - URR_{TDH}) \times TDR$$

$$Q_T = MAX(Q_T, 0)$$

$$URR_{DTH} = URR_{DTH} - Q_T$$

$$Q_N = MIN(URR_{DTH}, URR_{TDH}) \times NDR$$

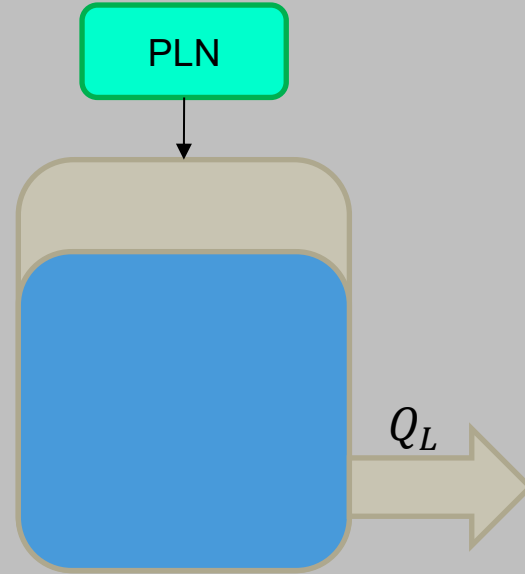
$$URR_{DTH} = URR_{DTH} - Q_N$$

$$PLN = URR_{DTH} \times ULC$$

$$URR_{DTH} = URR_{DTH} - PLN$$

HBV001a – Point Runoff Generation (Surface slow flow)

- Given PLN from URR
- Lower Reservoir (LRR)
 - Slower response (DRE)
 - Single outlet (linear)
 - Baseflow (Q_L)
 - Active all the time



HBV001a – Point Runoff Generation (Surface slow flow)

- Given PLN from URR
- Lower Reservoir (LRR)
 - Slower response (DRE)
 - Single outlet (linear)
 - Baseflow (Q_L)
 - Active all the time

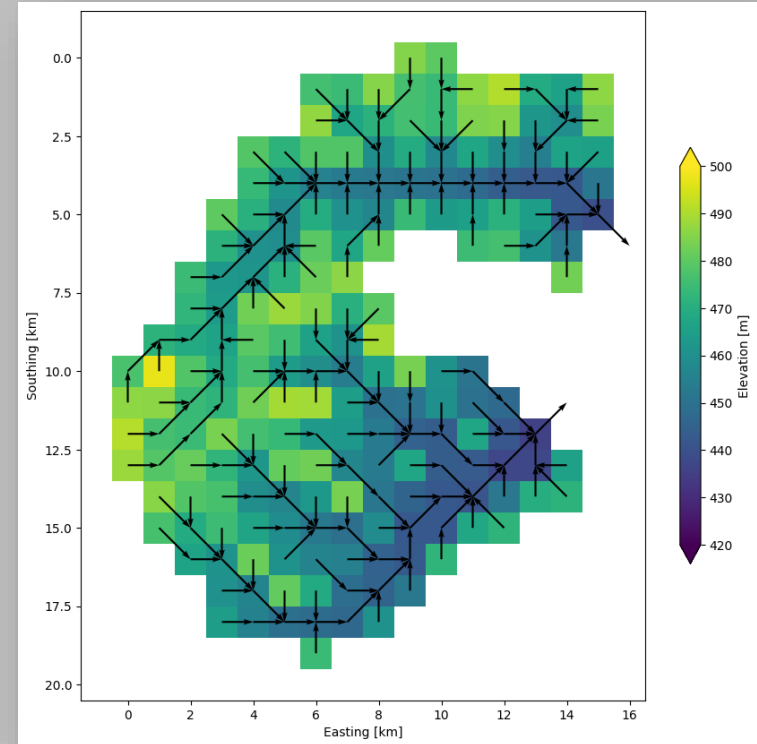
$$Q_L = LRR_{DTH} \times DRE$$

$$LRR_{DTH} = LRR_{DTH} - Q_L$$

$$LRR_{DTH} = LRR_{DTH} + PLN$$

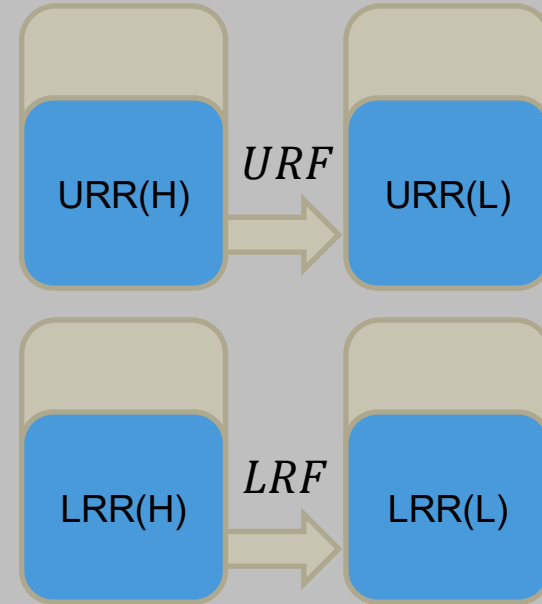
HBV001a – Point Runoff Generation (Sub-surface flows)

- D8 FDR for flow transfer
- Higher (H) to Lower (L) elevation
- Upper reservoir (URR)
 - Linear transfer to next cell (URF)
- Lower Reservoir (LRR)
 - Linear transfer to next cell (LRF)



HBV001a – Point Runoff Generation (Sub-surface flows)

- D8 FDR for flow transfer
- Higher (H) to Lower (L) elevation
- Upper reservoir (URR)
 - Linear transfer to next cell (URF)
- Lower Reservoir (LRR)
 - Linear transfer to next cell (LRF)



HBV001a – Point Runoff Generation (Sub-surface flows)

- D8 FDR for flow transfer
- Higher (H) to Lower (L) elevation
- Upper reservoir (URR)
 - Linear transfer to next cell (URF)
- Lower Reservoir (LRR)
 - Linear transfer to next cell (LRF)
- URR Update
 - Remaining RNF
 - Water split ratio (WSR)

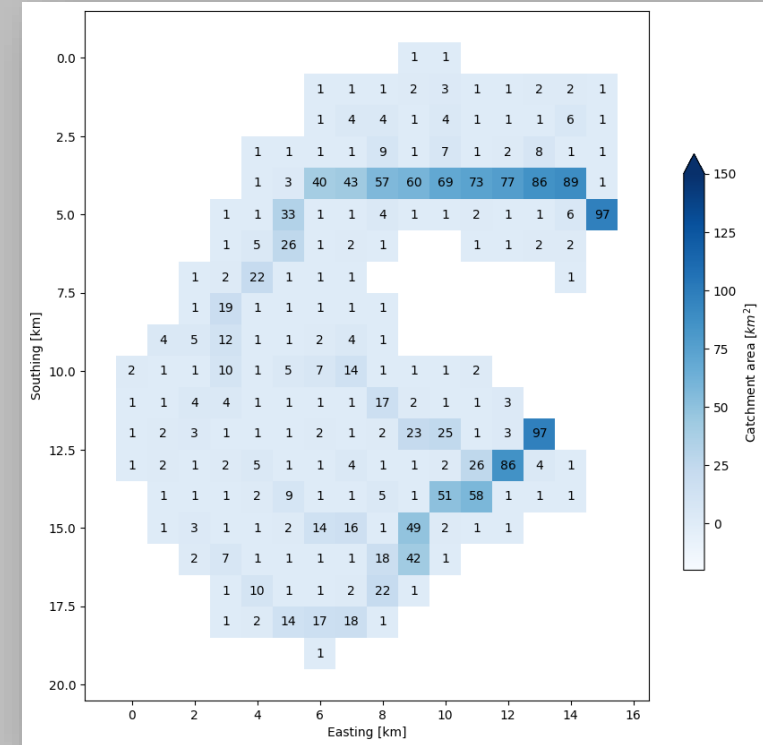
$$\begin{aligned}URF &= URR_{DTH}(H) \times UCT \\URR_{DTH}(H) &= URR_{DTH}(H) - URF \\URR_{DTH}(L) &= URR_{DTH}(L) + URF\end{aligned}$$

$$\begin{aligned}LRF &= LRR_{DTH}(H) \times LCT \\LRR_{DTH}(H) &= LRR_{DTH}(H) - LRF \\LRR_{DTH}(L) &= LRR_{DTH}(L) + LRF\end{aligned}$$

$$URR_{DTH} = URR_{DTH} + (WSR \times RNF)$$

HBV001a – Catchment Discharge

- Water leaves above and below surface
- Measured flow in river (Q_R)
 - Over surface (Q_P)
 - Sum of Q_T , Q_N , base Q_L flows
 - Flow depth in channel (Q_D)
 - Linear routed flow (Q_R)
 - Higher (H) to lower (L) elevations
- Flow under riverbed (URF and LRF)
 - Via porous medium
 - Riverbed and catchment interface



HBV001a – Catchment Discharge

- Water leaves above and below surface
- Measured flow in river (Q_R)
 - Over surface (Q_P)
 - Sum of Q_T , Q_N , base Q_L flows
 - Flow depth in channel (Q_D)
 - Linear routed flow (Q_R)
 - Higher (H) to lower (L) elevations
- Flow under riverbed (URF and LRF)
 - Via porous medium
 - Riverbed and catchment interface

$$Q_P = Q_T + Q_N + Q_L$$

$$Q_R = (Q_D + Q_P) \times CST$$

$$Q_D(H) = Q_D(H) - Q_R(H)$$

$$Q_D(L) = Q_D(L) + Q_R(L)$$

HBV001a – Mass Balance

- No implicit mass balance violation
- Account for mass at each step
- $Input(s) - Output(s) - \Delta Storage(s) = Zero$

That was it for today, thanks!